Work-Based Learning Module for

Computational Data Analytics

Year 2026

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WORK BASED LEARNING MODULE FOR BACHELOR OF SCIENCE (COMPUTATIONAL DATA ANALYTICS) WITH HONOURS

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Penerbit UTHM
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PREFACE

The Work-Based Learning (WBL) Module for the Bachelor of Science in Computational Data Analytics with Honours serves as a comprehensive guide to the implementation of WBL, providing clear and structured guidelines for students, academic supervisors, industry coordinators, and industry coaches involved in the one-year industrial training programme. Developed by the Computational Data Analytics panel from the Department of Mathematics and Statistics, Faculty of Applied Sciences and Technology, this module is designed to strengthen the connection between academic learning and real-world industry practices.

This module emphasizes the application of computational thinking, data analytics techniques, and professional practices in response to current and emerging industry demands. Its objectives are to enhance students' technical competencies, critical thinking, and problem-solving skills, while nurturing strong professional ethics and workplace readiness. In addition, the module aims to promote meaningful collaboration between academia and industry, ensuring mutual benefits for both students and host organizations. This module is intended to serve as an essential reference, inspire innovation and excellence, and support the development of graduates who are not only proficient in data analytics but also capable of making impactful contributions across diverse sectors in an increasingly data-driven world.

The Panel of Computational Data Analytics
Department of Mathematics and Statistics
Faculty of Applied Sciences and Technology

1.0 INTRODUCTION

1.1 THE DEFINITION OF WORK-BASED LEARNING

Work-Based Learning (WBL) is a structured educational approach that integrates academic study with professional practice in real-world settings. It allows students to translate theoretical knowledge acquired in the classroom into workplace applications, thereby enhancing both technical understanding and practical competency. WBL encompasses various formats, including internships, apprenticeships, cooperative education, or other supervised industry-based training experiences. Although no single definition of WBL has been universally adopted by scholars, its primary objective is to develop workplace readiness by equipping students with relevant experience, strengthening professional attributes, and facilitating a smooth transition from higher education to employment. Through authentic engagement with industry tasks, students acquire skills and perspectives that extend beyond traditional classroom learning and contribute meaningfully to their career development.

Figure 1 shows the correlation between students-industries based on the Edmund Model.

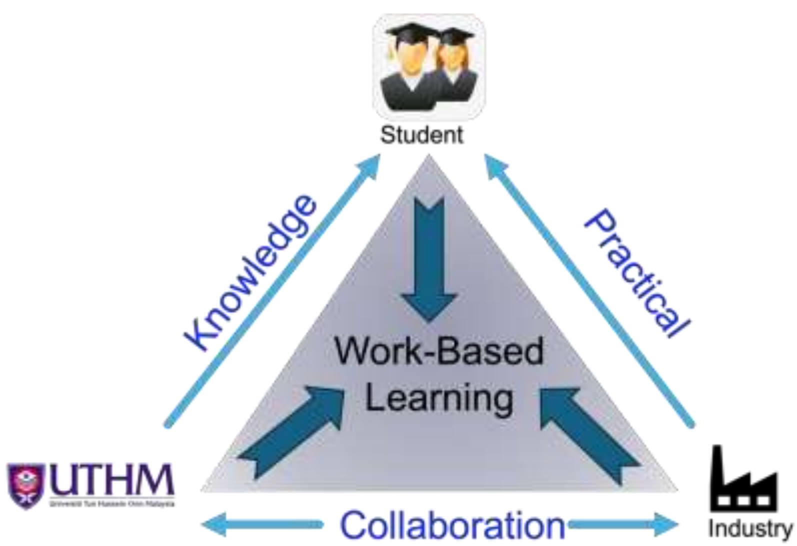

Figure 1 : Relationship between students-industries based on the Edmund Model.
(Source: *Garis Panduan Pembelajaran Berasaskan* Kerja (WBL) UTHM , 2021)

1.2 Bachelor of Science (Computational Data Analytics) with Honours

The Bachelor of Science (Computational Data Analytics) with Honours (BWS) program is tailored to develop graduates with strong computational, statistical, and analytical skills to meet the increasing demands of data-driven industries. The curriculum emphasizes both theory and application, preparing students to translate raw data into meaningful insights and solutions for real-world problems.

Offered under the 2u2i mode of study, the programme integrates two and a half years of structured academic coursework with one year of supervised industry placement. The curriculum equips students with fundamental and advanced knowledge in programming, statistics, machine learning, and big data technologies, which is subsequently applied in authentic professional and industrial contexts. Teaching and learning activities are delivered through a combination of lectures, laboratory sessions, workshops, industry projects, and seminars, supported by continuous and summative assessments to ensure the achievement of programme learning outcomes and holistic student development.

The Work-Based Learning (WBL) component is structured across three academic phases. In Semester 6, students undertake three industry-related courses, including the Industry Integrated Project Proposal, under the joint guidance of academic and industry supervisors. This is followed by Semester 7, in which students complete two courses focusing on the Industry Integrated Project Framework, project development, and final reporting, conducted within the host organization. Additionally, during the short semester, students enrol in one course on Applied Research Skills to support the preparation and refinement of their project work. Collectively, the WBL component carries a total of 25 credit hours, positioning it as a core element of the programme. Through formal partnerships with industry via MoUs and MoAs, students are placed in organizations that expose them to authentic data challenges and professional practices. The Industrial Final Year Project serves as the capstone experience, where students are expected to apply cutting-edge tools and methods in areas such as predictive analytics, optimization, machine learning, and business intelligence.

The BWS program is also accredited by the Institute of Analytics (IoA), ensuring that its curriculum meets international standards of excellence and professional relevance. By blending rigorous academic training with immersive industry experience, the BWS program ensures that graduates are industry-ready, adaptable, and capable of delivering data-driven solutions across diverse sectors including finance, healthcare, manufacturing, and technology.

The structure of the BWS program is illustrated in **Figure 2.**

Semester	Course Code	Course Name	Credit
1	UHB13102	English for General Communication	2
	UQI10102/ UQI10202	Islamic Studies / Moral Studies	2
	UQ*1**01	Co-Curriculum I	1
	UQU10903	Integrity and Anti-Corruption	3
	BWU10401	First Year Seminar	1
	BWS10103	Calculus	3
	BWS10203	Data Programming I	3
	BWS10303	Probability and Statistics	3
		Total Credit	**18**

Semester	Course Code	Course Name	Credit
2	UQI11202	Philosophy and Current Issues	2
	UHB1**02	Foreign Language	2
	UQU10702	Appreciation, Ethics and Civilization	2
	UQ*10001	Co-Curriculum II	1
	BWS10403	Fundamental to Data Science	3
	BWS10503	Discrete Mathematics	3
	BWS10603	Data Programming II	3
	BWS10703	Linear Algebra	3
		Total Credit	**19**

Semester	Course Code	Course Name	Credit
3	UHB23103	English for Technical Communication	3
	BWS20803	Regression Analysis	3
	BWS20903	Numerical Methods	3
	BWS21003	Inferential Statistics	3
	BWS21103	Data Structure and Algorithm	3
	BWS21203	Relational Database and Query Language	3
	BWU10102	Entrepreneuship	2
		Total Credit	**20**

Semester	Course Code	Course Name	Credit
4	BWU10202	Creativity and Innovation	2
	BWS21303	Applied Multivariate Analysis	3
	BWS21403	Non-Relational and NoSQL Database	3
	BWS21503	Special Topics on Computational Data Analytics	3
	BWS21603	Time Series Analysis	3
	BWS21703	Operational Research	3
		Total Credit	**17**

SHORT SEMESTER	BWS21803	Data Mining	3
	BWS21903	Exploratory Data Analysis & Visualization	3
		Total Credit	**6**

Semester	Course Code	Course Name	Credit
5	UHB33103	English for Professional Communication	3
	BWS32003	Text Analytics	3
	BWS32102	Risk Analysis	2
	BWS32203	Decision Support and Business Intelligence	3
	BWS33x03(1)	Elective I	3
	BWS33x03(2)	Elective II	3
	BWS33x03(3)	Elective III	3
		Total Credit	**20**

Semester	Course Code	Course Name	Credit
6	BWS44004	IT Professional Practice and Ethics *(WBL)	4
	BWS44104	Data Analytics and Management *(WBL)	4
	BWS44304	Industry Integrated Project Proposal	4
		Total Credit	**12**

SHORT SEMESTER	BWS44601	Applied Research Skills	1
		Total Credit	**1**

Semester	Course Code	Course Name	Credit
7	BWS44406	Industry Integrated Project Framework & Development*(WBL)	6
	BWS44506	Industry Integrated Project Report	6
		Total Credit	**12**

Figure 2: Curriculum Structure of Computational Data Analytics Program.

1.3 Aims and Learning Outcomes of Work-Based Learning (WBL)

In an increasingly competitive economy, higher education providers (HEPs) and industry partners play a critical role in developing graduates with high-level, industry-relevant skills. Work-Based Learning (WBL) strengthens this collaboration by integrating academic learning with authentic workplace experience, where employers contribute to programme design, learning delivery, student support, assessment, and, in some cases, programme resources. Such collaboration requires clear roles, quality assurance mechanisms, and appropriate staff development to ensure effective learning outcomes.

1.3.1 Aims of WBL

The aims of WBL describe the broad learning and teaching intentions of the programme and provide direction for curriculum design, quality assurance, and student learning experiences. Clear and explicit aims ensure that learners and stakeholders understand what the programme intends to achieve at the appropriate qualification level.

In the context of WBL, the primary aim is to enable students to learn in and from the workplace by developing industry-specific knowledge, skills, and professional competencies. WBL seeks to enhance students' employability, critical self-awareness, and personal development while supporting career progression. At the same time, WBL benefits employers through improved employee performance, innovation, and organisational productivity.

1.3.2 Learning Outcomes of WBL

Learning outcomes specify what students are expected to know, understand, and be able to do upon completion of WBL. These outcomes serve as benchmarks for academic standards, curriculum development, credit allocation, and assessment, and must align with programme and course learning outcomes as stipulated in the Malaysian Qualifications Framework (MQF).

WBL learning outcomes should be appropriate in scope, level, duration, and quality to ensure meaningful achievement. They should reflect MQF requirements across qualification levels, fields of study, and programme outcomes, and address the eight MQF learning outcome domains, namely:

i. Knowledge
ii. Practical skills
iii. Social skills and responsibilities
iv. Values, attitudes, and professionalism
v. Communication, leadership, and teamwork skills
vi. Problem-solving and scientific skills
vii. Information management and lifelong learning skills
viii. Managerial and entrepreneurial skills

Accordingly, WBL learning outcomes may include industry-specific competencies, employability and graduate skills, professional and ethical attributes, employer-required skills, achievement of workplace targets as evidence of competence, and career development goals identified through workplace evaluation and review.

2.0 IMPLEMENTATION OF WBL

2.1 Program Delivery

Work-Based Learning (WBL) delivery encompasses the approaches used to ensure that students achieve the prescribed learning outcomes. These include face-to-face instruction, online platforms, and blended learning. Within the academic setting, delivery is scheduled in accordance with the official timetable, whereas industry-based learning allows for greater flexibility, adapting to the nature and scope of the WBL course. The three core components of WBL delivery are presented in **Figure 3.**

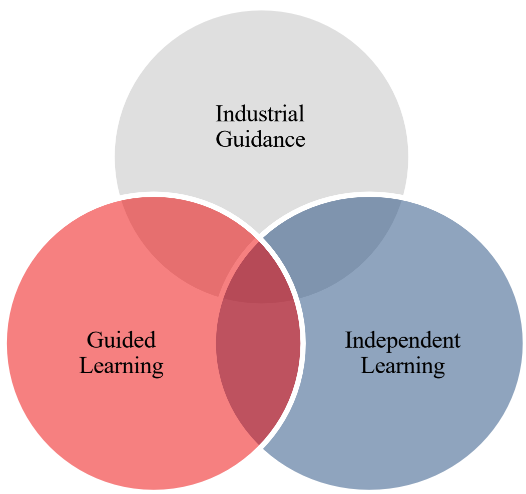

Figure 3: Course Delivery of WBL.

a) Industrial Guidance

Industrial guidance is provided by the Industry Coach (IC), who serves as a mentor to support students in understanding work tasks and related topics. Under this guidance, students gain hands-on workplace experience by applying theoretical knowledge to practical tasks. The effectiveness of this method relies on the expertise and experience of the IC, ensuring that students are able to meet the intended learning outcomes.

b) Guided Learning

Guided learning refers to structured learning activities conducted between lecturers and students, as specified in the official Class Schedule. Delivery follows the Academic Calendar and may take place face-to-face or online. Activities typically include presentations, discussions, question-and-answer sessions, dialogues, and elements of self-directed study.

c) Independent Learning

Independent learning takes place outside the formal class schedule and emphasizes the cultivation of lifelong learning skills. This approach requires students to be proactive and resourceful in expanding their knowledge beyond lecture materials. In the industry setting, this may involve completing additional assignments not specified by the IC. To fulfill such tasks, students are expected to independently source relevant information, deepen their understanding of topics, and engage in peer discussions or collaborative problem-solving.

2.2 Duration

The duration of the Work-Based Learning (WBL) program is one (1) year or 46 weeks, comprising two (2) regular semesters and one (1) short semester. Each regular semester spans 14 weeks, following the UTHM academic calendar for assessment purposes. In certain cases, the specific duration may be adjusted by the faculty.

2.2.1 WBL Academic Schedule

WBL academic schedule can be accessed through WBL official site.

2.2.2 Eligibility of WBL

a) Students are permitted to enrol in WBL after successfully completing all university courses. However, students may request consideration and approval from the Head of Department based on justifications that are subject to evaluation.

b) Students are not allowed to register for any other courses during the WBL period.

c) Students must attend the WBL orientation session conducted by the industry department before commencing their WBL placement.

2.3 Assessment Method

There are **TWO (2)** implementation modes for WBL involving teaching and supervision. Both implementations are carried out through three approaches, namely face-to-face (physical), online, or hybrid methods. Students are required to be physically present full-time throughout the WBL period at the organization/industry.

2.3.1 Implementation of Assessment Method

TWO (2) parties involved in WBL implementation which are Industry and University. Students will be assessed based on few criteria:

Industry	University
i. Weekly Report (WBL Progess Logbook Report) ii. Assessment Form (Industry Supervisor's Report During Observation/Practical) iii. Observation iv. Industrial Project Report	i. Weekly Report (WBL Progess Logbook Report) ii. Online Quiz iii. Assessment Form (University Supervisor's Report During Observation/Practical) iv. Video Presentation v. Industrial Project Report

2.4 Roles and Responsibilities

2.4.1 University (UTHM)

a) **WBL Coordinator (Faculty)**

i. Drafting the overall WBL plan after discussions with relevant implementers involved in the training program.

ii. Ensuring the successful implementation of the WBL according to the faculty's design.

iii. Ensuring that matters related to the implementation of the WBL are conducted in accordance with the relevant quality procedures and requirements.

iv. Developing and coordinating feedback forms for students, industry coach, and university supervisors for the purpose of Continuous Quality Improvement (CQI) in WBL.

b) **WBL Coordinator (Program)**

i. Assisting in the operational aspects of WBL program's management.

ii. Updating the list of students enrolled in WBL courses and identifying their respective fields.

iii. Updating the list of organizations/industries registered by students.

iv. Serving as an intermediary between university supervisors, industry coach and students.

v. Providing course code information, estimating the number of students, proposing dates, WBL duration, and training venue details to the department.

vi. Creating a schedule for University Supervision.

vii. Conducting briefings on WBL operations for university supervisors and students.

viii. Compiling activity reports for the overall implementation of WBL.

ix. Delivering briefings on instructional materials to instructors and university supervisors.

x. Collecting WBL assessment scores from industry coach.
xi. Maintaining records and reports related to WBL instruction.
xii. Facilitating course alignment sessions with industry coordinator and university supervisors.
xiii. Providing a Study Guide and relevant teaching documents.

c) **University Course Instructor**

i. Conducting teaching and learning activities based on modules and study guides.
ii. Monitoring the progress of students.
iii. Evaluating student compilation of project or task in final portfolio according to course requirement.

d) **University Supervisor**

i. Establishing meeting schedules, whether in-person or online according to the guidelines set by the program.
ii. Assessing the performance and progress of students' industrial project: proposal and final year project report.
iii. Coordinating, guiding, and providing mentorship to students undergoing WBL.
iv. Monitoring and identifying challenges faced by students in the organizational/industrial setting.

2.4.2 Industry

a) Industry Coordinator

The WBL Industry Coordinator acts as the Liaison Officer between the UTHM, students, academic staff, and Industry Coaches. Responsibilities include:

i. Identifying and selecting suitable Industry Coach(es).
ii. Assisting students in planning their placement at the industry.
iii. Resolving issues related to WBL students in the workplace.
iv. Maintaining regular and effective communication between university supervisors, industry coach(es), and students.

v. Coordinating the consistency of supervisory methods.
vi. Providing information to the faculty regarding the distribution list of Industry Coach(es) to students.
vii. Serving as an intermediary between the university and organizations/industries.
viii. Coordinating student files.
ix. Coordinating the implementation of teaching and learning (T&L) at the industry.
x. Coordinating the student evaluation process and submitting the reports to the UTHM within the stipulated time frame.

b) Industry Coach

i. Provide training and guidance to students according to the course requirements.
ii. Conduct course assessments and monitor students' progress.
iii. Guide students in their preparation for reports/assignments.
iv. Ensure that assessment reports are submitted to the Industry Coordinator within the stipulated time frame.
v. Ensure that students fully always comply with the safety and health regulations in the workplace.
vi. Review students' log books.
vii. Determine appropriate placements and task details for students.
viii. Provide instruction and guidance regarding assigned tasks.
ix. Evaluate students' performance using university-provided evaluation forms.
x. Submit student evaluations to the university within the specified timeframe.
xi. Inform the industry coordinator and faculty of any issues concerning students during their training in the organization/industry.
xii. Co-supervise the students' final year project course.

2.4.3 Student

i. Committed to executing every WBL activity during the organization/industry placement.
ii. Adhering to the confidentiality, safety regulations, and codes of conduct of the organization/industry.
iii. Completing all designated learning hours and assessments as outlined by the program.

3.0 ADMINISTRATION OF WBL

3.1 WBL Rules and Regulations

a) UTHM students undergoing WBL must obtain an offer letter that aligns with industry requirements before commencing their placement in the industry.

b) Students are required to consistently adhere to safety and health regulations established by the industry.

c) The industry is responsible for conducting orientation sessions.

d) The industry should provide placements, training on safety and health regulations, and furnish appropriate personal protective equipment facilities in accordance with the students' training needs.

e) UTHM students undergoing WBL are subject to rules and regulations as stipulated in the UTHM Bachelor's Academic Regulations Book.

f) UTHM students undergoing WBL are also subject to the rules and regulations stipulated by the industri.

3.1.1 Working Hours

The minimum working hours are 40 hours per week, while the maximum limit during students' Work-Based Learning (WBL) is set at 45 hours per week. Any duration exceeding this maximum limit requires mutual consent among the organization/industry, university, and the student.

3.1.2 Attendance

a) Students must ensure that their attendance is not less than **80%** throughout the WBL period. However, students are not allowed to be absent arbitrarily even if they have achieved 80% attendance.

b) Any absence beyond emergency reasons such as death, accidents, and the like must be reported to the organization/industry, and approval is subject to the decision of the organization/industry.
c) Students must also adhere to working hours and all industry regulations related to leave, sick leave, and public holidays, similar to other employees during the WBL period.
d) The industry should inform the WBL coordinator appointed for the purpose of necessary actions by UTHM regarding student absenteeism.

3.1.3 Relocation of WBL Placement

a) Students are not allowed to request a change in their placement without obtaining written approval from the UTHM WBL Coordinator.
b) Any request for a change in placement location must adhere to the existing terms and regulations governing UTHM's Industrial Training implementation.
c) UTHM is required to formally notify the relevant industry in writing regarding any changes in the placement location before permitting students to transition to another industry.
d) Students directed by the industry to work at a location other than the one previously approved before undergoing WBL must obtain permission from UTHM for this relocation. Approval will only be granted if the WBL
e) Coordinator is satisfied with the proposed new location to ensure the smooth progress of the students' WBL experience.

3.1.4 Termination of WBL

a) Students who fail to adhere to all rules and regulations of WBL, UTHM and industry/organization may be subject to disciplinary action.
b) If found guilty, students may face disciplinary measures, including warnings, fines, and training termination.
c) Repeat violations of discipline may result in the cessation of studies by the University Discipline Committee.

d) However, students have the option to request withdrawal from WBL with support from the industry to the WBL Coordinator at UTHM.
e) Any decision regarding the application to discontinue training will be considered based on the merits of each case.

3.1.5 Confidentiality

a) The regulations at the WBL site regarding attire, confidentiality, and office rules are subject to the terms set by the organization or industry.
b) Students are prohibited from disclosing any information (whether confidential or not) about the organization to external parties without the organization's permission.
c) Students are not allowed to print, copy, or take pictures of any documents or equipment deemed confidential without the organization's consent.
d) Students found to be in violation of any rules or regulations may be subjected to disciplinary actions or penalties based on:
e) Regulations that have been established by UTHM, and/or;
f) Regulations established by the industry, and/or;
g) Procedures for discipline follow relevant guidelines (Bahagian V Acara Tatatertib, Akta Institusi-institusi Pelajaran (Tatatertib) 1976) (Akta 174).
h) In the event of a conflict between UTHM regulations and industry regulations, UTHM regulations shall take precedence.

3.2 Insurance

Students will be covered by group insurance protection throughout their Work-Based Learning (WBL) period. However, industries are encouraged to obtain additional insurance coverage if deemed necessary.

3.3 Allowance and Facilities

a) The allowance during Work-Based Learning (WBL) is subject to the regulations of the organization/industry and is contingent upon an agreement between the organization/industry and the university.

b) The respective organization/industry is encouraged to refer to the Structured Internship Program Guidelines by Talent Corp (2019), which outlines the criteria enabling organizations/industries to qualify for double tax exemption.

c) The cost of living and student expenses throughout the WBL period are the responsibility of the student.

4.0 LIST OF COURSES

4.1 BWS44004: IT Professional Practice and Ethics

Synopsis:

This course discusses how these modern systems evolved over time and how the digitisation of data has changed science and business through the use of technologies aimed at supporting decision making. The course will look at the ethical implications of the widespread use of these systems and how governance frameworks are used to guide businesses in the ethical use of ICT.

Course Learning Outcomes (CLO):

At the end of this course, the students will be able to:

CLO1: Demonstrate and communicate a conceptual understanding of the implications of digital innovation within a business case study. (C4; LOD9; PLO6)

CLO2: Communicate a clear, coherent and independent analysis of the use of data analysis tools, Decision support Systems and IT governance and IT standards and frameworksin Business. (P4; LOD14; PLO10)

CLO3: Apply normative competency to understand and convey ethical implications in ICT artefact development. (A4; LOD16; PLO11, SC8)

Content Outline of the Course:

TOPIC 1: Digital Innovation and Disruption

Guidelines: Students will learn the fundamental concepts of ICT and digital transformation. They will explore historical and modern examples of digital disruption, understand how innovation reshapes industries, and evaluate its impact on organizations, customer behavior, and society.

TOPIC 2: Decision Support Systems and Analytics

Guidelines: Students will learn the concepts and applications of Decision Support Systems (DSS). They will differentiate between traditional and modern DSS approaches, including real-time analytics, and assess the effective use and potential misuse of DSS in business decision-making.

TOPIC 3: Governance, Audit, and Regulation of ICT

Guidelines: Students will learn the principles of ICT governance, audit, and regulation. They will understand the legal, ethical, and compliance requirements of cyberspace, explore governance frameworks, and evaluate practitioner insights into ensuring accountability and transparency.

TOPIC 4: Ethics, Professional Responsibility, and Society

Guidelines: Students will learn ethical analysis frameworks and professional codes of conduct in ICT. They will examine issues of system reliability, cybersecurity, and data protection, and analyze the wider societal impact of technology on trust, responsibility, and professional practice.

TOPIC 5: Management, Communication, and Team Dynamics

Guidelines: Students will learn essential management concepts and professional communication skills. They will apply teamwork and leadership principles, develop effective collaboration strategies, and reflect on current ICT trends to prepare for real-world workplace dynamics.

Teaching method: Work-based Learning/ Blended Learning

Materials to deliver the course: Lecture slides, case studies on ethical dilemmas in ICT, professional codes of ethics, company ICT governance policies, cybersecurity regulations, audit reports, practitioner interviews, and industry standards documentation.

Expected outcomes: Students will understand the foundations of professional ethics in ICT, critically analyze ethical, legal, and social issues in technology, and apply professional codes of conduct in workplace scenarios. They will be able to evaluate governance frameworks, ensure compliance with ICT regulations, and make ethically sound decisions in real-world professional practice. Students will also develop effective communication and teamwork skills necessary for ethical ICT management and decision-making.

Method of assessment: WBL Logbook / Case Study Analysis / Practical Demonstration / Online Quiz / Observation

Assessment:

PLO6: Digital Skills, **PLO10**: Entrepreneurial Skill, **PLO11**: Ethics and Professional, **SC8:** Normative Competency

Assessment	Assessor	CLO	PLO	Level	Assessment Tool	%
Continuous	University	CLO1	PLO6	C4	Online Quiz	10
		CLO1	PLO6	C4	(WBL Progess Logbook Report)	10
		CLO3	PLO11	A4-SC8	Video Presentation	10
	Industry	CLO1	PLO6	C4	(WBL Progess Logbook Report)	20
		CLO3	PLO11	A4-SC8	Observation	20
		CLO2	PLO10	P4	Demonstration	30
					Total	**100**

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4.2 BWS44104: Data Analytics and Management

Synopsis:

This course uses case studies to review the data challenges business confront and how data management and analytics is used to help make sound management decisions.

Course Learning Outcomes (CLO):

At the end of this module, the students will be able to:

CLO1: Recommed an appropriate model for different data types data process and analysis workflow computer systems for each organization; (C4; LOD2; PLO2)

CLO2: Interpret the business significance of the data, what it implies about the business, customers, etc (P4; LOD9; PLO6; SC5)

CLO3: Generating reports on the data, including appropriately constructed graphics and histograms that illustrate important features of the data.(A4; LOD13; PLO9)

Content Outline of the Course:

TOPIC 1: Introduction to Database Management Systems

Guidelines: Students will learn how to design, organize, and maintain databases using relational models, normalization, and referential integrity.

TOPIC 2: Data Preprocessing

Guidelines: Students will learn how to clean, prepare, and govern data for analysis, including reduction techniques and basic modeling.

TOPIC 3: Understanding Statistics

Guidelines: Students will learn basic statistical concepts such as data sets, size, range, and variation for data analysis.

TOPIC 4: Business Intelligence and Analytics

Guidelines: Students will learn to apply descriptive statistics, visualize data, and explore datasets to generate useful insights.

TOPIC 5: Managerial Decision-Making

Guidelines: Students will learn how to use data queries, models, and reports to support managerial decisions.

Teaching Method: Work-based Learning

Materials to deliver the course: Lecture slides and notes, Company's data management procedures and governance policies, Case studies and real industry datasets, Data analytics standards or compliance documents.

Expected outcomes:

Students will be able to analyze datasets using appropriate techniques, perform data management and analytics tasks confidently with suitable tools, and demonstrate professional values by adhering to governance and ethical standards in practice.

Method of assessment: WBL Logbook /Case Study Analysis / Video Presentation, Practical Observation/ Online Quiz

Assessment:

PLO2: Cognitive Skill, **PLO6:** Digital Skill, **PLO9:** Personal Skill, **SC5:** Collaboration Competency

Assessment	Assessor	CLO	PLO	Level	Assessment Tool	%
Continuous	University	CLO1	PLO2	C4	Online Quiz	10
		CLO1	PLO2	C4	(WBL Progess Logbook Report)	10
		CLO3	PLO9	A4	Video Presentation	10
	Industry	CLO1	PLO2	C4	(WBL Progess Logbook Report)	30
		CLO2	PLO6	P4-SC5	Demonstration	20
		CLO3	PLO9	A4	Observation	20
					Total	**100**
					Total	**100**

References:

1. Rick Sherman. (2014). Business Intelligence Guidebook - From Data Integration to Analytics, Morgan Kaufmann, ISBN 978-0124114616
2. Liebowitz, Jay. (2013). Big data and business analytics. (HD38.7.B53 2013)
3. Huang, Mao Lin and Huang, Weidong. (2014). Innovative approaches of data visualization and visual analytics. Hershey, PA : Information Science Reference (QA76.9.I52 .I56 2014)
4. Runkler, Thomas A. (2014). Data analytics : models and algorithms for intelligent data analysis. New York : Springer Vieweg (QA76.9.D343 .R86 2012)
5. Das, Subrata. (2014). Computational business analytics. Boca Raton, FL : CRC (HF5548.2 .D38 2014)

4.3 BWS44304: Industry Integrated Project Proposal

Synopsis:

This course would provide the essential steps to be taken in the process of producing quality in writing project proposal for the data science field. This course focusses on starting with selecting the title, writing abstract and introduction, selection of appropriate methodology by using existing or developing new methodology and writing the reference. This course also provides student how to manage the timetable and how to overcome the project challenges along the study. All skills learned are transferable and applicable to other genre writing as well.

Course Learning Outcomes (CLO):

At the end of this module, the students will be able to:

CLO1: Propose viable solution to any data analytics applications at related industry or agency. (C5; LOD8; PLO5)

CLO2: Produce a professional, competitive, compelling and successful project proposal. (P5; LOD9; PLO6)

CLO3: Manage appropriate resources including data collection in preparing the proposal. (A4; LOD16; PLO11)

Content Outline of the Course:

TOPIC 1: Introduction to Integrated Project Proposal

TOPIC 2: Background Knowledge and Context Area

- **i.** Review on current practices

TOPIC 3: Consideration of Proposed Methods

- **i.** Methods approach to conduct Integrated Project
- **ii.** Design project to get the best results/findings

TOPIC 4: Strategy, Timetable and Project Challenges

- i. Stage of project
- ii. Gantt chart
- iii. Challenges encounter/overcome

TOPIC 5: Sources and Referencing

- **i.** List of key references
- **ii.** Other relevant material related to reference

Teaching Method: Final Year Project

Materials to deliver the course: Lecture Slide, templates and samples for proposal writing, project work, practical session, case studies, meeting with university supervisor, etc.

Expected outcomes:

Students will be able to prepare a comprehensive integrated project proposal by defining the title, objectives, hypotheses, research questions, and scope; select appropriate methods; plan the project timeline and anticipate challenges; and compile relevant references and supporting materials.

Assessment method: Proposal/ Presentation/ Reflective Log

Assessment:

PLO5: Communication Skill, **PLO6:** Digital Skills, **PLO11:** Ethics and Professionalism

Assessment	Assessor	CLO	PLO	Level	Assessment Tool	%
Continuous	University	CLO1	PLO5	C5	Proposal Project Report (SV)	15
		CLO1	PLO5	C5	WBL Logbook (SV)-Progress Report	10
		CLO1	PLO5	C5	Proposal Project Report (EX)	10
		CLO3	PLO11	A4	Presentation (EX)	10
	Industry	CLO1	PLO5	C5	Proposal Project Report	20
		CLO2	PLO6	P5	Demonstration	20
		CLO3	PLO11	A4	Observation	15
					Total	**100**

References:

1. Buku Log Latihan Industri UTHM. (Bahagian A). Pejabat Penerbit UTHM
2. Guidelines to good practices: work-based learning (GGP:WBL) MQA 2015
3. Panduan Penulisan Tesis UTHM
4. Mohd Noor, N. 2011. Writing Research And Thesis Proposals : Guidelines And Examples.

Shah Alam : University Publication Centre (UPENA) (LB2369 .N66 2011)
5. Sharma, S. C. (2006). Introductory Operation Research. New Delhi: Discovery Publishing House (T57.6 .S524 2006)

BWS44406: Industry Integrated Project Framework and Development

Synopsis:

This course will expose the students on the process of conducting research in order to provide the skills and ability in carrying out research project framework and development in the data analytical field. The covered areas consisting of research design, project management, design of innovation, data collection and analysis. All the works can consider to publish via conference and journal publication.

Course Learning Outcomes (CLO):

At the end of this module, the students will be able to:

CLO1: Describe a solution to data analytical problems using a systematic research methodology effectively. (C5; LOD3; PLO2)

CLO2: Use appropriate and relevant resources to develop a project based on organization needs. (P5; LOD13; PLO9)

CLO3: Practise good ethics in managing the project. (A4; LOD18; PLO11)

Content Outline of the Course:

TOPIC 1: Research Design

i. Determine the type of research/convention selected to conduct the study.
ii. Sequence the different stages and link them with data collection and outputs.
iii. Evaluate the strengths and weaknesses of construction methods.

TOPIC 2: Construction Innovation

i. Learn how to handle an integrated project based on the chosen project type.

ii. Explore types of innovations related to current industry practices.

iii. Assess the value of creative solutions.

iv. Understand basic innovation management principles.

TOPIC 3: Project Delivery Methods

i. Understand the fundamentals of project delivery methods.

ii. Apply design-build methods to implement projects.

iii. Select appropriate delivery methods for different project types.

TOPIC 4: Data Collection and Analysis

i. Collect and manage primary and secondary quantitative data sets.

ii. Analyze data, interpret findings, and validate results.

TOPIC 4: Attributes of Framework

i.Understand requirements for conferences related to the project.

ii. Prepare materials suitable for publication and professional dissemination

Method of assessment: Final Year Project

Materials to deliver the course: Lecture Slide, Gantt charts and project scheduling tools, project work, practical session, case studies, meeting with university supervisor, etc.

Expected outcomes:

Students should be able to develop and execute project or research approaches, design effective research projects, collect and analyze data proficiently, and understand and apply the attributes of relevant frameworks to their work.

Assessment method: Interview / Progress Report/ Presentation / Demonstration / Practical Task/ Reflective Log

Assessment

PLO2: Cognitive Skill**, PLO9:** Personal Skill**, PLO11:** Ethics and Professionalism

Assessment	Assessor	CLO	PLO	Level	Assessment Tool	%
Continuous	University	CLO1	PLO2	C5	WBL Logbook Progress Report	20
		CLO3	PLO11	A5	Video Presentation	25
	Industry	CLO1	PLO2	C5	WBL Logbook Progress Report	15
		CLO2	PLO9	P5	Observation	40
					Total	**100**

References:

1. Buku Log Latihan Industri UTHM. (Bahagian A). Pejabat Penerbit UTHM
2. Guidelines to good practices: work-based learning (GGP:WBL) MQA 2015
3. Panduan Penulisan Tesis UTHM
4. Mohd Noor, N. 2011. Writing Research And Thesis Proposals : Guidelines And Examples. Shah Alam : University Publication Centre (UPENA) (LB2369 .N66 2011)
5. Gupta, A. K. 2013. Design and Analysis of Experiments. World Scientific, Singapore. Call number: QA279 .G86 2013.
6. Antony, J. 2003. Design of Experiments for Engineers and Scientists. Butterworth-Heinemann, Oxford. Call number: QA279 .A57 2003.

4.4 BWS44506: Industry Integrated Project Report

Synopsis:

This course provides conceptual framework of project report and professional communication. Students will have opportunity to complete integrated project into report and thesis, preparing and delivering presentations of report successfully and to implementation of this delivery method on an industrial project.

Course Learning Outcomes (CLO):

At the end of this module, the students will be able to:

CLO1: Propose the innovation in integrated project report clearly. (C5; LOD4; PLO2)

CLO2: Engage in continuous learning to apply integrated problem-solving competency by using relevant frameworks and developing inclusive, equitable solutions for complex sustainable development problems in data analytics and related fields. (P5; LOD13; PLO9,SC6)

CLO3: Manage a good communication skills while presenting the project. (A4; LOD16; PLO11)

Content Outline of the Course:

TOPIC 1: Report and Technical Paper Write Up

i. Fine tuning Chapter 1, 2 and 3 of the research proposal
ii. Interpret research data
iii. Write a standard format of Industry Integrated Project and technical paper

TOPIC 2: Presentation

i. Present the results of the research in oral presentation session
ii. Question and answer session

TOPIC 3: Submission of Final Report And Technical Paper

Teaching Method: Final Year Project

Materials to deliver the course: Slide, Gantt charts and project scheduling tools, project work, practical session, case studies, meeting with university supervisor, etc.

Expected outcomes:

Students will be able to compile a comprehensive industry final report by documenting project objectives, methodology, data collection and analysis, findings, and conclusions, while demonstrating the ability to reflect critically on their work, apply academic knowledge to real industry contexts, and present results professionally.

Assessment method: Project / Industrial Product / Progress Report/ Demonstration / Project/ Progress Report

Assessment

PLO2: Cognitive Skill, **PLO9:** Personal Skill, **PLO11:** Ethics & Professionalism, **SC6:** Integrated Problem Solving Competency

Assessment	Assessor	CLO	PLO	Level	Assessment Tool	%
Continuous	University (SV)	CLO1	PLO2	C5	WBL Logbook Progress Report	5
		CLO1	PLO2	C5	Project Report	10
	University (EX)	CLO1	PLO2	C5	Project Report	10
		CLO2	PLO9	P5	Project Report	10
		CLO3	PLO11	A4	Presentation	10
	Industry	CLO1	PLO2	C5	Project Report	20
		CLO2	PLO9-SC6	P5	Demonstration	25
		CLO1	PLO2	C5	WBL Logbook Progress Report	10
Total						**100**

References:

1. Buku Log Latihan Industri UTHM. (Bahagian A). Pejabat Penerbit UTHM
2. Guidelines to good practices: work-based learning (GGP:WBL) MQA 2015
3. Panduan Penulisan Tesis UTHM
4. Mohd Noor, N. 2011. Writing Research And Thesis Proposals : Guidelines And Examples. Shah Alam : University Publication Centre (UPENA) (LB2369 .N66 2011)
5. Gupta, A. K. 2013. Design and Analysis of Experiments. World Scientific, Singapore. Call number: QA279 .G86 2013.

4.5 BWS44601: Applied Research Skills

Synopsis:

This course provides students with guided opportunities to strengthen their research progress between the Research Proposal and Research Framework and Report phases. Emphasis is given to improving and updating their approved proposal, developing detailed research methodology, and initiating report writing with proper academic style and referencing. Through short seminars, peer discussions, and supervisor feedback, students will refine their research design, ensure methodological consistency, and enhance academic writing quality. This course bridges proposal planning with research implementation, fostering readiness for the final research phase.

Course Learning Outcomes (CLO):

At the end of this module, the students will be able to:

CLO1: Apply and integrate knowledge to solve real-world problems with critical self-reflection, and refine a research proposal based on feedback and methodological needs. (C4; LOD6; PLO4; SC7)

CLO2: Apply appropriate research methodology in developing the research framework. (P4; LOD11; PLO8)

CLO3: Produce a preliminary research report section with ethical considerations that demonstrates proper academic writing, citation, and structure. (A4; LOD16; PLO11)

Content Outline of the Course:

TOPIC 1: 1.Refinement Summary

1.1 Overview of proposal feedback

1.2 Identifying improvements

TOPIC 2: 2. Methodology Enhancement Workshop

2.1 Improvements to the research design and instruments

2.2 Explanation of alignment between objectives, research questions, and methods

TOPIC 3: Academic Writing Clinic

3.1 Structure of research chapters, paraphrasing, citation tools

TOPIC 4: Research Design Consistency Matrix

4.1 Alignment between objectives, research questions, variables, methods, and data requirements**.**

TOPIC 5: Preliminary Analysis Plan

5.1 Initial plan for data analysis

5.2 Sample tables/graphs using dummy data to illustrate expected outputs

TOPIC 6: Progress Seminar & Reflection

6.1 Presentation of progress and research readiness for next phase

Teaching Method: Final Year Project

Materials to deliver the course: Slide, Gantt charts and project scheduling tools, project work, practical session, case studies, meeting with university supervisor, seminar, etc.

Expected outcomes:
Upon successful completion of this course, students will be able to critically review and refine their approved research proposal to ensure clarity, coherence, and alignment with the intended research objectives and framework; develop a detailed, systematic, and methodologically sound research methodology appropriate to the study; ensure consistency between the research problem, objectives, methodology, and proposed analysis in preparation for research implementation; produce initial sections of a research report using proper academic writing conventions and correct referencing style; integrate constructive feedback from supervisors and peers to enhance research quality; and effectively communicate research progress in academic settings such as seminars and peer discussions using appropriate scholarly language.

Assessment method: Project / Industrial Product / Progress Report/ Demonstration / Project/ Progress Report

Assessment

PLO4: Interpersonal Skill, **PLO8:** Leadership, Autonomy and Responsibility, **PLO11:** Ethics & Professionalism, **SC7:** Self Awareness Competency

Assessment	Assessor	CLO	PLO	Level	Assessment Tool	%
Continuous	University (SV)	CLO1	PLO4-SC7	C4	WBL Logbook- Progress Report	10
		CLO1	PLO4	C4	Project Refinement Report	20
		CLO2	PLO8	P4	Demonstration	10
		CLO3	PLO11	A4	Video Presentation	5
	Industry	CLO1	PLO4-SC7	C4	WBL Logbook Progress Report	25
		CLO2	PLO8	P4	Observation	30
Total						**100**

References:

1. Buku Log Latihan Industri UTHM. (Bahagian A). Pejabat Penerbit UTHM

2. Guidelines to good practices: work-based learning (GGP:WBL) MQA 2015

3. Panduan Penulisan Tesis UTHM

4. Mohd Noor, N. 2011. Writing Research And Thesis Proposals : Guidelines And Examples. Shah Alam : University Publication Centre (UPENA) (LB2369 .N66 2011)

5. A. Yavuz Oruc.,Handbook of scientific proposal writing, Boca Raton : CRC Press , 2012.

WBL PROGRESS REPORT LOGBOOK

Semester/Session	
Student Name	
Matrix No.	
Faculty Name	
Company Name	

Instruction:

i. Students must complete this report **weekly**.
ii. This WBL logbook encompasses the evaluation of **all courses** in the respective semester. Please ensure that all activities/tasks conducted in the industry are properly recorded in this WBL logbook.
iii. This logbook must be completed in accordance with the **requirements of courses** BWS44004, BWS44104, and BWS44304.
iv. The completed logbook must be uploaded to **AUTHOR every Friday** (weekly submission).
v. The **Industry Coach (IC)** will verify the WBL Logbook **once (1) per month**. Therefore, students are required to **compile four (4) weeks** of logbook entries for monthly verification.
vi. The verified monthly logbook (after IC verification) must also be uploaded on the **fourth Friday of each month.**
vii. Students are encouraged to include **supporting materials** such as diagrams, photos, charts, technical drawings, or any relevant evidence to strengthen and support the reported activities.
viii. Students must ensure that **no confidential, sensitive, or proprietary company information** is disclosed in the logbook. Any restricted documents, internal data, or classified materials must not be uploaded without prior permission from the company.

MATHEMATICS AND STATISTICS DEPARTMENT

FACULTY OF APPLIED SCIENCES AND TECHNOLOGY
COMPUTATIONAL DATA ANALYTICS PROGRAM

Date/week: ____

Project/Activities
BWS44004:   **BWS44104:**   **BWS44304:**

Accomplishments
BWS44004:   **BWS44104:**   **BWS44304:**

Work in Progress
BWS44004:   **BWS44104:**   **BWS44304:**

WEEKLY WORK ACCOMPLISHMENT SUMMARY
APPENDIX

For Industry Use

Criteria	Very Unsatisfied	Unsatisfied	Satisfied	Good	Very good
Records of activities in the logbook					
Organization and structure					
Quality of Work					
Planning and Organization					
Attitude					

Verified by Industry Coach:

Name:

Date:

Company:

Official Stamp:

REFERENCE

No	Item	Level	Very Unsatisfied	Unsatisfied	Satisfied	Good	Very Good
1.	Logbook (Content)	C4	Activities, reflections, and outcomes are mostly lacking from entries.	Activities, reflections, and outcomes are incompletely described in entries.	Activities, reflections, and outcomes are detailed in entries.	Most actions, reflections, and consequences are detailed in entries.	Activity, reflection, and outcome descriptions are detailed.
2.	Organization and structure	C3	Logbook is disorganized with no clear structure.	Logbook is poorly organized with a structure that is difficult to follow.	Logbook is generally organized but may lack a clear structure in some areas.	Logbook is mostly organized with a clear structure.	Logbook is well-organized with a logical structure that is easy to follow.
3.	Quality of Work	A3	Fails to produce quality work independently without attention to detail or standards.	Rarely creates quality work autonomously, lacking detail and standards.	Occasionally performs quality work independently with attention to detail and standards.	Regularly delivers high-quality, detailed work independently as per criteria.	Independently generates high-quality work with attention to detail and high standards.
4.	Planning and Organization	A4	Unplanned, unstructured work, and poor resource and time management.	Plan and arrange work little, setting few goals and milestones and straining to manage resources and time.	Sets goals and milestones, manage resources and time wisely, and plans work.	Plan and organize work, create goals and milestones, and manage resources and time.	Plans and organizes work exceptionally well, setting clear goals and milestones, and efficiently managing resources and time.
5.	Attitude	P3	Displays negative attitude, uncooperative, resistant to feedback, and lacks professionalism.	Occasionally shows negative attitude and requires frequent reminders to behave professionally.	Generally acceptable attitude but inconsistent in professionalism and responsiveness.	Maintains positive attitude, cooperative, receptive to feedback, and behaves professionally.	Consistently demonstrates excellent attitude, highly respectful, proactive, mature, and sets a positive example to others.

5.2 Rubric for Industry Assessment WBL Logbook

Student's Name:
University Supervisor's Name:
Company:
Date:

Instructions: This booklet will be used for every 4 weeks submission appraisal. Please tick (/) in the score column.

No	Item	Level	Very Poor (1)	Poor (2)	Fair (3)	Good (4)	Very Good (5)	Weightage	Score
1.	Logbook (Content)	C4	Activities, reflections, and outcomes are mostly lacking from entries.	Activities, reflections, and outcomes are incompletely described in entries.	Activities, reflections, and outcomes are detailed in entries.	Most actions, reflections, and consequences are detailed in entries.	Activity, reflection, and outcome descriptions are detailed.	2	12
2.	Organization and structure	C3	Logbook is disorganized with no clear structure.	Logbook is poorly organized with a structure that is difficult to follow.	Logbook is generally organized but may lack a clear structure in some areas.	Logbook is mostly organized with a clear structure.	Logbook is well-organized with a logical structure that is easy to follow.	0.8	8
3.	Quality of Work	A3	Fails to produce quality work independently without attention to detail or standards.	Rarely creates quality work autonomously, lacking detail and standards.	Occasionally performs quality work independently with attention to detail and standards.	Regularly delivers high-quality, detailed work independently as per criteria.	Independently generates high-quality work with attention to detail and high standards.	0.8	4
4.	Planning and Organization	A4	Unplanned, unstructured work, and poor resource and time management.	Plan and arrange work little, setting few goals and milestones and straining to manage resources and time.	Sets goals and milestones, manage resources and time wisely, and plans work.	Plan and organize work, create goals and milestones, and manage resources and time.	Plans and organizes work exceptionally well, setting clear goals and milestones, and efficiently managing resources and time.	0.6	3
5.	Attitude	P3	Displays negative attitude, uncooperative, resistant to feedback, and lacks professionalism.	Occasionally shows negative attitude and requires frequent reminders to behave professionally.	Generally acceptable attitude but inconsistent in professionalism and responsiveness.	Maintains positive attitude, cooperative, receptive to feedback, and behaves professionally.	Consistently demonstrates excellent attitude, highly respectful, proactive, mature, and sets a positive example to others.	**0.6**	**3**
TOTAL									**30**

5.3 Rubric for University Assessment Logbook

Student's Name:
University Supervisor's Name:
Company:
Date:

Instructions: This booklet will be used for every weeks' submission appraisal. Please tick (/) in the score column.

BWS44004									
No	**Item**	**Level**	**Very Poor (1)**	**Poor (2)**	**Fair (3)**	**Good (4)**	**Very Good (5)**	**Weightage**	**Score**
1	Conceptual Understanding of Digital Innovation	C3	Does not understand digital innovation concepts. Explanation is incorrect or irrelevant to the topic.	Shows minimal understanding. Concepts are mentioned but inaccurately explained or confused.	Demonstrates basic understanding of digital innovation concepts but lacks depth or clarity.	Clearly explains key digital innovation concepts accurately and appropriately.	Demonstrates deep, critical and well-integrated understanding of digital innovation concepts with strong theoretical grounding.	2	10
2	Analysis of Business Implications	C4	No analysis of how digital innovation affects the business case.	Limited or superficial discussion of business implications.	Identifies some business implications but analysis lacks depth or critical evaluation.	Provides logical and relevant analysis of digital innovation impacts on the business case.	Provides comprehensive, critical and insightful evaluation of short- and long-term business implications.	3	15
3	Clarity and Coherence of Communication	C4	Communication is unclear, disorganized, and difficult to understand.	Ideas are poorly structured with weak logical flow.	Communication is understandable but lacks strong organization and coherence.	Ideas are clearly structured, logically presented, and easy to follow.	Communication is highly professional, coherent, persuasive, and academically well-articulated.	2	10
								TOTAL	**35**

BWS44104									
No	**Item**	**Level**	**Very Poor (1)**	**Poor (2)**	**Fair (3)**	**Good (4)**	**Very Good (5)**	**Weightage**	**Score**
1	Understanding of Data Types & Processing	C3	Demonstrates no understanding of data types and processing workflow.	Limited understanding with major inaccuracies.	Basic understanding but explanation lacks depth.	Good understanding of data types and processing workflow.	Excellent and comprehensive understanding of structured/unstructured data, workflow, and system integration.	2	10
2	Ability to Recommend Appropriate Model	C4	Unable to recommend a suitable model. Recommendation is incorrect or irrelevant.	Recommendation is weak and not suitable for the data type or organization.	Recommendation is partially appropriate but lacks strong justification.	Recommends an appropriate model aligned with data type and workflow with clear justification.	Recommends highly appropriate and optimized model tailored to data type, processing workflow, and organizational needs with strong technical justification.	3	15
3	Data Interpretation and Decision Support	C5	Fails to interpret analytical results meaningfully, does not link findings to decision-making, and contains no reflective evaluation.	Demonstrates weak interpretation of results; linkage to decision-making is unclear or incorrect; and reflection is superficial.	Demonstrates basic interpretation of results with limited linkage to decision-making; explanations are general; and reflection is simple.	Demonstrates accurate interpretation of analytical results with relevant contextual explanation; links findings to decision-making with minor gaps; and includes meaningful reflection on outcomes.	Demonstrates insightful interpretation of analytical results, clearly links data findings to business or organizational decision-making, evaluates implications and limitations of the results, and reflects critically on the effectiveness of data-driven decisions.	2	10
								TOTAL	**35**

BWS44306									
No	**Item**	**Level**	**Very Poor (1)**	**Poor (2)**	**Fair (3)**	**Good (4)**	**Very Good (5)**	**Weightage**	**Score**
1.	Research Background & Problem Statement	C4	Problem is unclear, irrelevant, or not related to industry/agency context.	Problem is weakly defined with minimal industry relevance.	Problem is generally defined but lacks strong justification or supporting evidence.	Problem is clearly defined, relevant to industry, and supported with logical justification.	Problem is critically analysed, strongly justified with clear industry significance and supporting evidence.	1.0	5
2.	Quality and Viability of Proposed Solution	C4	Proposed solution is inappropriate, unrealistic, or not related to the problem.	Solution is weak, lacks feasibility, and poorly aligned with data analytics application.	Solution is relevant but lacks depth, feasibility analysis, or technical clarity.	Solution is appropriate, feasible, and clearly aligned with data analytics techniques and industry needs.	Solution is highly viable, technically sound, innovative, and critically justified with strong alignment to real industry constraints and requirements.	1.0	5
3	Clarity of Objectives and Analytical Approach	C5	Objectives are unclear or not measurable. No clear analytical direction.	Objectives are vague and weakly connected to the proposed solution.	Objectives are generally clear but lack precision or measurable indicators.	Objectives are clear, measurable, and logically connected to the proposed analytical approach.	Objectives are precise, measurable, strategically structured, and strongly aligned with the proposed data analytics methodology.	1.0	5
4	Expected Outcomes and Impact	C5	Expected outcomes are unclear or unrealistic.	Outcomes are weakly described and lack industry relevance.	Outcomes are stated but lack depth or impact analysis.	Outcomes are realistic, relevant, and clearly beneficial to the industry/agency.	Outcomes are critically evaluated, measurable, sustainable, and demonstrate strong potential impact on organizational performance.	1.0	5
								TOTAL	**20**

BWS44404									
No	**Item**	**Level**	**Very Poor (1)**	**Poor (2)**	**Fair (3)**	**Good (4)**	**Very Good (5)**	**Weightage**	**Score**
1.	Understanding of Data Analytical Problem	C2	Fails to identify the problem; shows little to no understanding of the data analytical challenge.	Problem identification is unclear or partially incorrect; limited understanding of the data analytical challenge.	Identifies the problem superficially; understanding of the data analytical challenge is moderate.	Identifies the problem and its context with minor gaps; shows good understanding of the data analytical challenge.	Clearly identifies the problem, its context, and significance; demonstrates deep understanding of the data analytical challenge.	1.0	5
2	Application of Systematic Research Methodology	C3	Fails to apply systematic research methodology; steps are incorrect or missing.	Attempts to apply methodology but major errors present; steps are poorly justified.	Applies methodology partially; some steps are unclear or justification is weak.	Applies methodology appropriately with minor errors; provides mostly clear justification of steps.	Applies systematic research methodology accurately and comprehensively; justifies steps clearly with appropriate techniques.	2.0	10
3	Data Analysis and Interpretation	C4	Fails to perform analysis or interpret results; no connection to the problem or solution.	Analysis is incomplete or contains errors; interpretation is weak; limited connection to the problem.	Performs basic analysis; interpretation is partial or superficial; some connection to the problem is made.	Performs mostly accurate analysis; interprets results reasonably well; links findings to the problem with minor gaps.	Performs thorough and accurate data analysis; interprets results insightfully; clearly links findings to the problem and proposed solution.	2.0	10
4	Clarity and Coherence of Proposed Solution	C5	Proposed solution is confusing, illogical, or absent.	Solution is unclear, poorly structured, or not aligned with methodology.	Solution is somewhat understandable but lacks logical flow or clarity; partially aligned with methodology.	Solution is mostly clear and structured; minor inconsistencies with methodology or problem context.	Proposed solution is logical, well-structured, and clearly explained; aligns with the research methodology and problem context.	2.0	10
								TOTAL	**35**

BWS44506									
No	**Item**	**Level**	**Very Poor (1)**	**Poor (2)**	**Fair (3)**	**Good (4)**	**Very Good (5)**	**Weightage**	**Score**
1.	Analysis, Interpretation, and Synthesis of Findings	C4	No meaningful analysis or interpretation of findings.	Minimal or unclear analysis and interpretation of findings.	Basic analysis and interpretation of findings with limited synthesis.	Accurate analysis and clear interpretation of findings with appropriate linkage to objectives.	Critical analysis and insightful interpretation of findings, synthesising results with relevant literature and clearly relating them to research objectives and framework.	1.0	5
2.	Academic Reasoning, Reflection, and Report Development	C5	Strong academic reasoning, deep reflection on research progress, clear identification of limitations, and continuous improvement of the project report structure and content.	Superficial reflection and weak academic reasoning.	Basic reflection and acceptable academic reasoning with limited improvement to the report.	Sound academic reasoning and meaningful reflection with evidence of report improvement.	Strong academic reasoning, deep reflection on research progress, clear identification of limitations, and continuous improvement of the project report structure and content.	2.0	10
								TOTAL	**15**

BWS44601									
No	**Item**	**Level**	**Very Poor (1)**	**Poor (2)**	**Fair (3)**	**Good (4)**	**Very Good (5)**	**Weightage**	**Score**
1.	Research Thinking and Conceptual Understanding	C3	Little or no understanding of research concepts or research thinking.	Limited understanding of research concepts with weak or unclear reasoning.	Basic understanding of research concepts with acceptable but limited explanation.	Clear understanding of research concepts with sound reasoning and minor gaps in explanation.	Critical and in-depth understanding of research concepts, clearly explaining research problems, objectives, and theoretical foundations with strong analytical reasoning.	0.8	4
2.	Analytical Reflection and Research Problem-Solving	C4	No meaningful reflection or problem-solving.	Superficial reflection with unclear problems or solutions.	Research progress with basic problem identification and solutions.	Meaningfully on research challenges and proposes appropriate solutions.	Critical reflection on research progress, identifies challenges accurately, analyses causes, and proposes well-reasoned improvement strategies.	1.2	6
								TOTAL	**10**

5.4 Rubric for Industry Assessment (Observation & Demonstration Evaluation)

Student's Name:
Industrial Supervisor's Name:
Company:
Date:

Instructions: This booklet will be used for mid-semester and end-semester appraisal. Please tick (/) in the score column.

BWS44004 IT Professional Practice and Ethics									
1	Able to plan and organise tasks assigned professionally.	P2						1	5
2	Able to explore, investigate and engage in project activities with curiosity and responsibility.	P3						3	15
3	Able to follow organisational rules, policies, confidentiality guidelines, and ICT usage ethics.	P4						4	20
4	Able to demonstrate ethical judgement in handling data, information, and digital artefacts.	A4-SC8						**TOTAL**	**40**
5	Able to adapt to different roles, situations, and work expectations in the organisation.	A3-SC8						0.6	3
6	Able to respect, accept and respond appropriately to the opinions and feedback of others.	A2-SC8						0.4	2
7	Able to manage emotions, demonstrate self-control, and maintain professional behaviour.	A4-SC8						1	5
								TOTAL	**10**
8	Able to understand and apply fundamental workplace etiquette, governance and ethical principles.	A3-SC8						1	5
9	Able to demonstrate leadership or initiative when required (leading self and/or others).	A4-SC8						3	15
								TOTAL	**20**
BWS44104 Data Analytics and Management									
1	Able to apply analytical skills to interpret data or solve workplace-related tasks.	P3						4	20
2	Able to use tools/software/devices related to analysis, reporting or data handling with care and proficiency.	P4						3	15
3	Facilitate collaborative problem-solving by generating simple findings, summaries, or reports that support workplace or project-based decision-making.	P4-SC5						1	5
								TOTAL	**40**
4	Demonstrate the ability to collaborate effectively with peers and relevant stakeholders in managing and organising workplace documents, datasets, or digital resources in a responsible manner.	P4-SC5						2	10
								TOTAL	**10**
5	Apply appropriate strategies to utilise available resources effectively, including time, tools, data, and systems, to complete assigned tasks efficiently and productively.	A4						3	15
6	Exhibit professional behaviour, positive work attitude, and commitment when working collaboratively in project tasks and workplace-related activities.	A4						1	5
								TOTAL	**20**
BWS44304: Industry Integrated Project Proposal									
1	Operate and utilise digital tools, software applications, and data-handling technologies proficiently for analysis, reporting, and information management tasks.	P3						4	20

2	Organise, manage, and maintain digital documents, datasets, and information resources systematically and securely in accordance with workplace practices.	P4						3	15
3	Select and use digital resources efficiently, including time-management tools, analytical platforms, data systems, and supporting technologies, to complete assigned tasks within given constraints.	P4						1	5
								TOTAL	**40**
4	Demonstrate responsible digital work practices, professional conduct, and commitment when performing technology-enabled tasks and project activities.	A4						2	10
								TOTAL	**10**
6	Produce basic digital outputs, such as data summaries, visual reports, or analytical findings, that support operational or managerial decision-making in the workplace.	A3						1	5
								TOTAL	5
BWS44406: Industry Integrated Project Framework and Development & BWS44506: Industry Integrated Project Report									
1	Demonstrates professional ethics and discipline by adhering to organisational policies, project guidelines, and workplace standards throughout the execution of the industry-integrated project	P3						1	5
2	Shows responsibility and accountability in managing assigned project tasks, meeting agreed milestones, and completing deliverables within the specified timeframe.	P4						3	15
3	Demonstrates self-management and professional maturity by planning work independently, managing time effectively, handling project responsibilities responsibly, and maintaining a consistent commitment to quality and continuous improvement throughout the industry-integrated project.	P5-SC6						4	20
								TOTAL	**40**
4	Displays initiative, adaptability, and positive work attitude by responding appropriately to feedback, resolving project-related challenges, and continuously improving personal performance during project implementation.	A3						2	10
								TOTAL	**10**
5	Practices professionalism and ethical judgement by making responsible decisions, managing project-related issues with integrity, and contributing positively to the overall project and workplace environment.	A4						3	15
6	Exhibits effective interpersonal and teamwork skills by collaborating constructively with industry supervisors, project team members, and stakeholders during project planning and development activities.	A4						1	5
								TOTAL	**5**
BWS44601: Applied Research Skills									
1	Apply appropriate research methodologies competently and consistently to develop a structured and coherent research framework aligned with the research objectives.	P3						4	20
2	Select, adapt, and justify suitable research methods and techniques autonomously in constructing the research framework based on the research context.	P4						3	15
3	Lead the development and refinement of the research framework by integrating methodological decisions, theoretical grounding, and practical considerations.	P4						1	5
								TOTAL	**40**
4	Demonstrate responsibility and independence in making informed methodological decisions during the development of the research framework.	A3						2	10
								TOTAL	**10**
5	Exhibit professional integrity and ethical responsibility in applying research methodologies, including transparency, rigor, and accountability.	A4						3	15

6	Demonstrate leadership, initiative, and commitment to scholarly standards when guiding research framework development and responding constructively to feedback.	A4						1	5
								TOTAL	**5**

5.5 Rubric for Video Presentation Assessment

Student's Name:
University Supervisor's Name:
Company:
Date:

Instructions: This booklet will be used by end of semester submission appraisal. Please tick (/) in the score column.

No	Item	Level	Very Poor (1)	Poor (2)	Fair (3)	Good (4)	Very Good (5)	Weightage	Score
1	Technical proficiency with digital tools	A3	Demonstrate no technical proficiency with digital tools	Show limited technical proficiency with digital tools	Displays basic technical proficiency with digital tools	Shows good technical proficiency with digital tools	Demonstrates excellent technical proficiency with digital tools	1	5
2	Able to integrate the content relevantly related to data analytics and management/ ICT professional practice and ethics	A2	Fail to integrate the content.	Show limited integration of the content relevantly	Occasionally produces satisfactory integration of the relevant content.	Show good integration of the content relevantly	Demonstrate excellent integration the content relevantly	0.4	2
3	Normative Competency – Understanding and Conveying Ethical Implications in ICT Artefact Development (BWS44004)	A4	Clearly identifies ethical issues in ICT artefact development; effectively conveys implications using precise reasoning; demonstrates deep understanding of ethical principles.	Identifies most ethical issues; conveys implications clearly; shows good understanding of ethical principles.	Identifies some ethical issues; conveys implications partially; understanding of ethical principles is basic.	Identifies few ethical issues; conveys implications unclearly; limited understanding of ethical principles.	Fails to identify ethical issues; cannot convey implications; shows little to no understanding of ethical principles.	0.6	3
4	Data Reporting – Generating Report (BWS44104)	A4	Produces clear, accurate reports; uses appropriately constructed graphics and histograms that effectively illustrate key data features; insights are well explained.	Produces mostly accurate reports; uses graphics and histograms appropriately; insights are mostly clear.	Produces basic reports; some graphics/histograms used but may be inaccurate or unclear; insights partially explained.	Produces incomplete or unclear reports; graphics/histograms poorly constructed; insights weak or confusing.	Fails to produce reports; no graphics or histograms; no insights provided.	0.6	3
								TOTAL	**10**

5.6 Rubric for Project Proposal Report

Student's Name:
University Supervisor's Name:
Company:
Date:

Instructions: Supervisor Evaluation Rubric - Report (10%)

No	Item	Level	Very Poor (1)	Poor (2)	Fair (3)	Good (4)	Very Good (5)	Weightage	Score
1	Introduction	C4	● Consist of information regarding project background and **ONE** of the following ● Project justification ● Preliminary finding. ● Flow and organization which are very poorly stated.	● Consist of information regarding project background and **ONE** of the following: • Project justification. • Preliminary finding. ● Flow and organization which are poorly stated.	● Consist of information regarding project background and **TWO** of the following: • Project justification. • Preliminary finding. ● Flow and organization which are fairly stated.	● Consist of information regarding project background and the following: • Project justification. • Preliminary finding. • Flow and the organization have a well-organized structure.	• Consist of information regarding project background and the following: • Project justification. • Preliminary finding. • Flow and organization are presented in a precise and well-organized structure.	0.6	3
2	Problem Statement	C5	Very poorly structured regarding the research field's existing problem with no reference citations and unorganized structure.	• Poor structure regarding the research field's existing problem with no reference citations and unorganized structure. • Flow and organization are presented in an unorganized structure.	• Fair structure regarding the existing problem of the research field with reference citations. • Flow and organization are presented in an organized structure.	• Clear and well-structured regarding the existing problem of the research field with reference citations. • Flow and organization are presented in an organized structure.	• Very clear and well-structured regarding the existing problem of the research field, complete with reference citations. • Flow and organization are presented in a precise and well-organized structure.	3	12

3	Objectives and Scope of Study	C4	The objectives and scopes consist of **ONE** of the following: • Purpose. • Well-defined parameter. ● Presented in an organized structure.	The objectives and scopes consist of **TWO** of the following: ● Purpose. ● Well-defined parameter. ● Presented in an organized structure.	The objectives and scopes consist of **THREE** of the following: ● Purpose. ● Well-defined parameter. ● Presented in an organized structure.	The objectives and scopes consist of the following: ● Purpose. ● Well-defined parameter. ● Presented in an organized structure.	The objectives and scopes consist of the following: ● Purpose. ● Well-defined parameter. ● Presented in a precise and well-organized structure.	0.6	3
4	Literature Review	C4	Consist of relevant information regarding previous work, fundamental knowledge, and the preliminary theories which: ● Not referred from any indexed publications ● References are not correctly quoted. ● identified the work contribution poorly. ● Flow and the organization presented in an unorganized structure.	Consist of relevant information regarding previous work, fundamental knowledge, and the preliminary theories which: ● Referred from some of the indexed publications. ● References are not correctly quoted. ● identified the work contribution poorly. ● Flow and the organization presented in an unorganized structure.	Consist of relevant information regarding previous work, fundamental knowledge, and the preliminary theories which: • Referred from some of the indexed publications. • References are quoted correctly. • identified the work contribution. ● Flow and organization are presented in an organized structure.	Consist of relevant information regarding previous work, fundamental knowledge, and the preliminary theories which: • Referred from fully indexed publications. • References are quoted correctly. • identified the work contribution. ● Flow and organization presented in a well-organized structure.	Consist of relevant information regarding previous work, fundamental knowledge, and the preliminary theories which: • Referred from fully indexed publications. • References are quoted correctly. • identified the work contribution. • Flow and the organization presented in a precise and well-organized structure.	0.8	4
5	Research Methodology	C3	Consist of detailed information regarding workflow,	Consist of detailed information regarding workflow,	Consist of detailed information regarding workflow,	Consist of detailed information regarding workflow,	Consist of detailed information regarding workflow,	1.6	8

			strategy, and approach and the following: • Very poor in defining the method or tool used. • Flow and the organization presented and not in an organized structure.	strategy, and approach and the following: • Insufficient in defining the method or tool used. • Flow and the organization presented and not in an organized structure.	strategy, and approach and the following: • Reasonably defined the method or tool used. • Flow and organization is presented in an organized structure.	strategy, and approach and the following: • Good in defining the method or tool used. • Flow and organization presented in a well-organized structure.	strategy, and approach and the following: • Very good in defining the method or tool used. • Flow and the organization presented in a precise and well-organized structure.		
6	Planning (Gantt Chart and Report Structure)	C2	No Gantt chart is available.	The Gantt chart is not well organized.	Follow the schedule according to the plan. The Gantt chart is presented.	Follow the schedule according to the plan. The Gantt chart is presented and organized.	Follow the schedule according to the plan. The Gantt chart is well-presented and organized.	0.6	3
7	Conclusion	C2	The conclusions are very poorly stated.	Conclusions are made following the overall process, including **ONE** of the following: • The main goal. • Method. • Expected results are correctly stated.	Conclusions are made following the overall process, including **TWO** of the following: • The main goal. • Method. • Expected results are correctly stated.	Conclusions are made following the overall process, including: • The main goal. • Method. • Expected results are presented in an orderly manner without ignoring the element of sustainability.	Conclusions are made following the overall process, including: • The main goal. • Method. • Expected results are creatively presented in a precise and orderly manner without ignoring the element of sustainability.	0.6	3
8	References	C1	• Poor formatting. No references cited are indexed publications.	• 50% according to format. • Less than 30% of references cited are	• 80% according to format. • Less than 31 - 60% of references cited are indexed publications.	• 100% according to format. • Less than 61 - 80% of references cited are indexed publications.	• 100% according to format. • Well-presented and organized. • More than 81% of references cited are indexed publications.	0.2	1

				indexed publications.					
9	Format (Writing, Citation, References, Table and Figure)	C1	Does not follow the prescribed format at all.	Mostly does not follow the prescribed format.	Follows the prescribed format very loosely.	Follows the prescribed format closely.	Follows the prescribed format very closely.	0.2	1

5.7 **Rubric for Integrated Project Presentation**

Student's Name:
Examiner Industry/ University's Name:
Company:
Date:

Instructions: Examiner Evaluation Rubric - Presentation (30%)

No	Item	PLO	Level	Very Poor (1)	Poor (2)	Fair (3)	Good (4)	Very Good (5)	Weightage	Score
1	Project Understanding		A2	Deliver the presentation with any **ONE** of the following: • Well-organized ideas. • Clear and precise explanation using simple sentences. • Good technical knowledge. • Ability to explain the subject material from a different perspective.	Deliver the presentation with any **TWO** of the following: • Well-organized ideas. • Clear and precise explanation using simple sentences. • Good technical knowledge. • Ability to explain the subject material from different perspectives.	Deliver the presentation with any **THREE** of the following: • Well-organized ideas. • Clear and precise explanation using simple sentences. • Good technical knowledge. • Ability to explain the subject material from different perspectives.	Deliver the presentation with the following: • Well-organized ideas. • Clear and precise explanation using simple sentences. • Good technical knowledge. • Ability to explain the subject material from different perspectives.	Deliver the presentation with: • Very well-organized ideas. • Very clear and precise explanations using simple sentences. • Very good technical knowledge. • Ability to explain the subject material from different perspectives.	0.6	3
2	Presentation Style (Appearance and Delivery)		A4	Deliver the presentation with any **ONE** of the following: • Good fluency in English • Proper attire. • Good communication skills i.e. loud and clear. • Creative and well-organized	Deliver the presentation with any **TWO** of the following: • Good fluency in English • Proper attire. • Good communication skills i.e. loud and clear. • Creative and well-organized	Deliver the presentation with any **THREE** of the following: • Good fluency in English • Proper attire. • Good communication skills i.e. loud and clear. • Creative and well-organized	Deliver the presentation with the following: • Good fluency in English • Proper attire. • Good communication skills i.e. loud and clear. • Creative and	Deliver the presentation with: • Very good fluency in English • Proper attire. • Very good communication skills i.e. loud and clear. • Very creative and well-organized	2	10

				slide presentation.	slide presentation.	slide presentation.	well-organized slide presentation.	slide presentation.		
3	Question and Answer Session		A2	Unable to answer **ALL** questions.	Able to answer **SOME** questions with any of the following: • Precisely with clear information but not in detail. • Without hesitation. • Immediately. • In different perspectives.	Able to answer **SOME** questions with: • Precisely with clear information but not in • detail. • Without hesitation. • Immediately. • In different perspectives.	Able to answer **ALL** questions with: • Precisely with clear information but not in detail. • Without hesitation. • Immediately. • In different perspectives.	Able to answer **ALL** questions with: • Precisely with clear information in detail. • Without hesitation. • Immediately. • In different perspectives.	0.4	2
4	Self Confidence		A3	• Reading from slides. • No eye contact. • Shaky voice. • No body gestures.	• Less confident. • Very little eye contact. • Less body gestures. • Has the tendency to read from slides.	• Confident. • Occasional eye contact. • Few body gestures.	• Highly confident. • Frequent eye contact. • With • appropriate body gesture.	• Very highly confident. • Full eye contact. • With appropriate body gestures.	2	10
5	Impact on Industrial Technology Solution for Sustainable Development		A4	Poor understanding of the impact of the project on society, economy, and environment.	Understand the impact of the project on any **ONE** of the following: • Society. • Economy. • Environment.	Understand the impact of the project on any **TWO** of the following: • Society. • Economy. • Environment.	Understand the impact of the project on society, the economy, and the environment.	Very clear understanding of the impact of the project on society, economy, and environment.	1	5
								TOTAL		

5.8 Rubric for Thesis/Final Report (Examiner)

Student's Name:
University Supervisor's Name:
Company:
Date:

Instructions: Supervisor Evaluation Rubric - Thesis (10%)

No	Item	PLO	Level	Very Poor (1)	Poor (2)	Fair (3)	Good (4)	Very Good (5)	Weightage	Score
1	Abstract		C4	Consist of a brief introduction, objective, and **ONE** of the following: ● Method. ● Result. ● Conclusion ● Presented in a well-organized structure.	Consist of a brief introduction, objective, and **TWO** of the following: ● Method. ● Result. ● Conclusion ● Presented in a well-organized structure.	Consist of a brief introduction, objective, and **THREE** of the following: ● Method. ● Result. ● Conclusion ● Presented in a well-organized structure.	Consist of a brief introduction, objective, and the following: ● Method. ● Result. ● Conclusion. ● Presented in a well-organized structure.	Consist of a brief introduction, objective, and the following: ● Method. ● Result. ● Conclusion. ● Presented in a well-organized structure.	0.8	4
2	Introduction, Objectives, and Scope of Study		C4	Consist of further information regarding the problem statement, objectives clarifications, and **ONE** of the following • Project justification. • Scope. • Flow and organization which are correctly stated.	Consist of further information regarding the problem statement, objectives clarifications, and **TWO** of the following • Project justification. • Scope. • Flow and organization which are correctly stated.	Consist of further information regarding the problem statement, objectives clarifications, and **THREE** of the following • Project justification. • Scope. • Flow and organization which are correctly stated.	Consist of further information regarding the problem statement, objectives clarifications, and the following • Project justification. • Scope. • Flow and organization are presented in a well-organized structure.	Consist of further information regarding the problem statement, objectives clarifications, and the following: • Project justification. • Scope. • Flow and organization presented in a precise and well-organized structure.	1	5
3	Literature Review		C4	Consist of relevant information regarding previous work,	Consist of relevant information regarding previous work,	Consist of relevant information regarding previous work, fundamental	Consist of relevant information regarding previous work,	Consist of relevant information regarding previous work, fundamental knowledge, and	0.6	3

				fundamental knowledge, and the preliminary theories which: • Not referred from any indexed publications. • References are not correctly quoted. • identified the work contribution poorly. • Flow and organization are presented in an unorganized structure.	fundamental knowledge, and the preliminary theories which: • Referred from some of the indexed publications • References are not correctly quoted. • identified the work contribution poorly. • Flow and organization are presented in an unorganized structure.	knowledge, and the preliminary theories which: • Referred from some of the indexed publications. • References are quoted correctly. • identified the work contribution. • Flow and organization is presented in an organized structure.	fundamental knowledge, and the preliminary theories which: • Referred from fully indexed publication. • References are quoted correctly. • identified the work contribution. • Flow and organization are presented in a well-organized structure.	the preliminary theories which: • Referred from fully indexed publications. • References are quoted correctly. • identified the work contribution. • Flow and organization are presented in a precise and well-organized structure.		
4	Research Methodology		C3	Consist of detailed information regarding workflow, strategy, and approach and the following: • Very poor in defining the method or tool used. • Flow and organization presented and not in an organized structure.	Consist of detailed information regarding workflow, strategy, and approach and the following: • Insufficient in defining the method or tool used. • Flow and organization presented and not in an organized structure.	Consist of detailed information regarding workflow, strategy, and approach and the following: • Reasonably defined the method or tool used. • Flow and organization presented in an organized structure.	Consist of detailed information regarding workflow, strategy, and approach and the following: • Good in defining the method or tool used. • Flow and organization are presented in a well-organized structure.	Consist of detailed information regarding workflow, strategy, and approach and the following: • Very good in defining the method or tool used. • Flow and the organization presented in a precise and well- organized structure.	2	10
5	Results		C5	The results are flawed and not in accordance with the theory.	The results are not fully in accordance with the theory.	• The results are in accordance with the theory.	• The results are in accordance with the theory. • Presented and elaborated in a	• The results are in accordance with the theory. • Systematically presented and elaborated in a	1.4	7

						• Presented and elaborated in an orderly manner.	precise and orderly manner.	precise and orderly manner.		
6	Analysis and Discussion		C5	Results are discussed with: • Appropriate theory. • Deplorable analytical and critical manners.	Results are discussed with: • Appropriate theory. • Flawed comparison with similar work done. • The analysis is poorly presented.	Results are discussed with: • Appropriate theory. • Comparison with similar work done. • The analysis is presented in an orderly manner.	Results are discussed with: • Appropriate theory. • Comparison with similar work done. • The analysis is presented in a precise and orderly manner.	Results are discussed with: • Appropriate theory. • Comparison with similar work done. • High level of the analysis presented in a precise and orderly manner.	1.6	8
7	Conclusion and Recommendation		C2	Conclusions are very poorly stated and recommendations are irrelevant.	Conclusions are made in accordance with the analysis of results and poorly fulfill all the objectives and recommendations.	Conclusions are made in accordance with the analysis of results and reasonably fulfill the objectives and recommendations.	• Conclusions are made in accordance with the analysis of results and consist of findings that contribute to others. • Recommendations for future works are discussed in brief.	• Conclusions are made in accordance with the analysis of results and consist of detailed findings that contribute to others. • Recommendations for future works are to be done.	0.8	4
8	References		C1	• Poor in formatting. • No references cited are indexed publications.	• 50% according to format. • Less than 30% of references cited are indexed publications.	• 80% according to format. • Less than 31 - 60% of references cited are indexed publications	• 100% according to format. • Less than 61 -80% of references cited are indexed publications.	• 100% according to format. • Well-presented and organized. • More than 81% of references cited are	1	5

								indexed publications		
9	Fulfillment of Research Objectives		C2	According to Conclusion and Recommendation.	According to Conclusion and Recommendation.	According to the Conclusion and Recommendatio n on.	According to Conclusion and Recommendation.	According to the Conclusion and Recommendatio n.	0.8	4
10	Format (Writing, Citation, References, Table and Figure)		A4	Does not follow the prescribed format at all.	Mostly does not follow the prescribed format	Follows the prescribed format very loosely.	Follows the prescribed format closely.	Follows the prescribed format very closely.	2	10
								TOTAL		60

PROJECT TITLE

STUDENT'S NAME
MATRIC NO.

A proposal submitted in partial
fulfillment of the requirement for the award of the
Degree of Bachelor of Science (Computational Data Analytics) with Honour

Faculty of Applied Sciences and Technology
Universiti Tun Hussein Onn Malaysia

MONTH 202X

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